



Rail Vehicle Suspensions and Components

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B O S T O N



Overview

- Rail Car Types.
- Rail Car Suspensions: Why are suspensions important; reasons for suspension design choices.
- Rail Car Truck and Suspension Components.



Railway Car Types

- Industry started simple: only a few car types carried everything.
- As builders and railroads looked for efficiencies, new car types were developed.
- Freight cars and passenger cars began to diverge in design
- 2 axles lead to 4 axles
- More passenger space
- Better comfort
- Higher speeds
- Specialized passenger car makers grew as passenger rail service expanded throughout the world

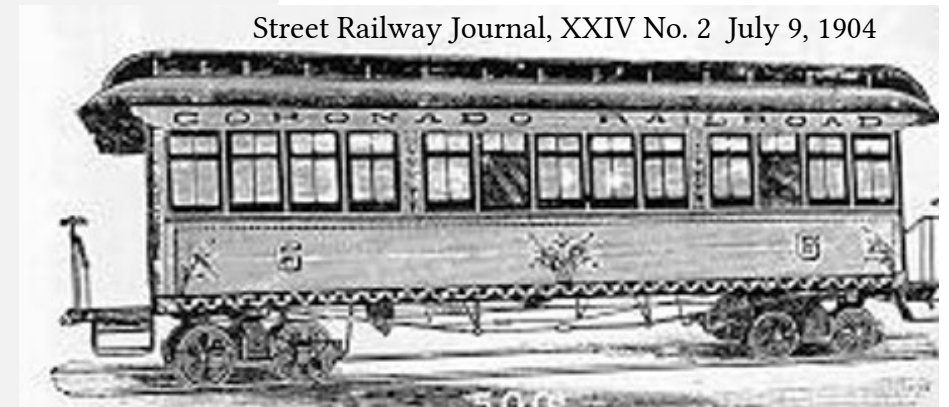
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Street Railway Journal, XXIV No. 2 July 9, 1904



Passenger Car Types

- Since the start of the railway, large variety of designs available
- Often custom or “one-off” designs, unique to a city or a route within a city.
- Urban and intercity; streetcars to complete trainsets.
- Light to heavy; low speed to very fast.





Passenger Car Weights and Measures

- Intercity passenger coach car from ~60t AW0 to ~75t AW3
- 5-car streetcar AW0 is ~50t; AW3 is about 65t
- Ratio of ~1.3 : 1 between empty weight and loaded weight

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Passenger Car Weights and Measures

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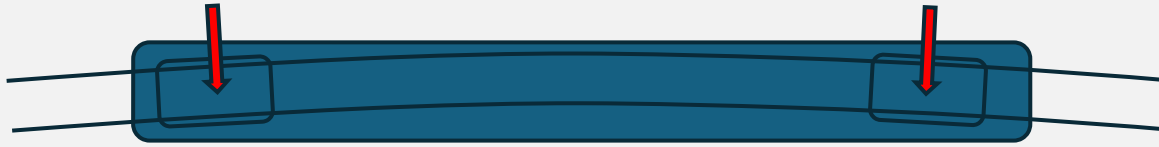
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- Intercity: up to 160mph, curves as sharp as ~720 feet
- Modern Streetcar: Curves as sharp as 36 feet, up to ~40 mph

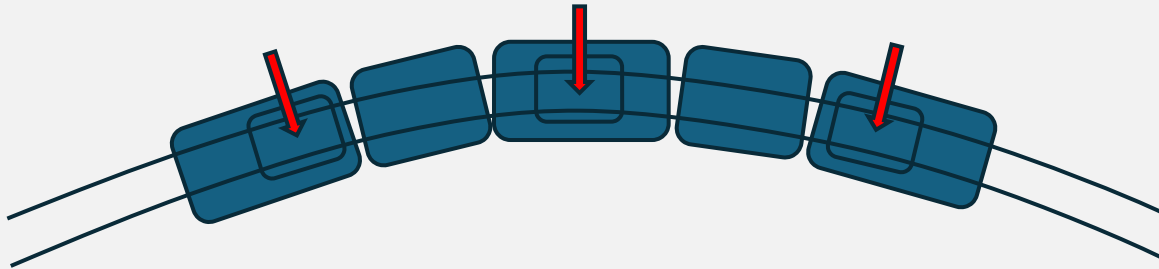


Passenger Car Weights and Measures

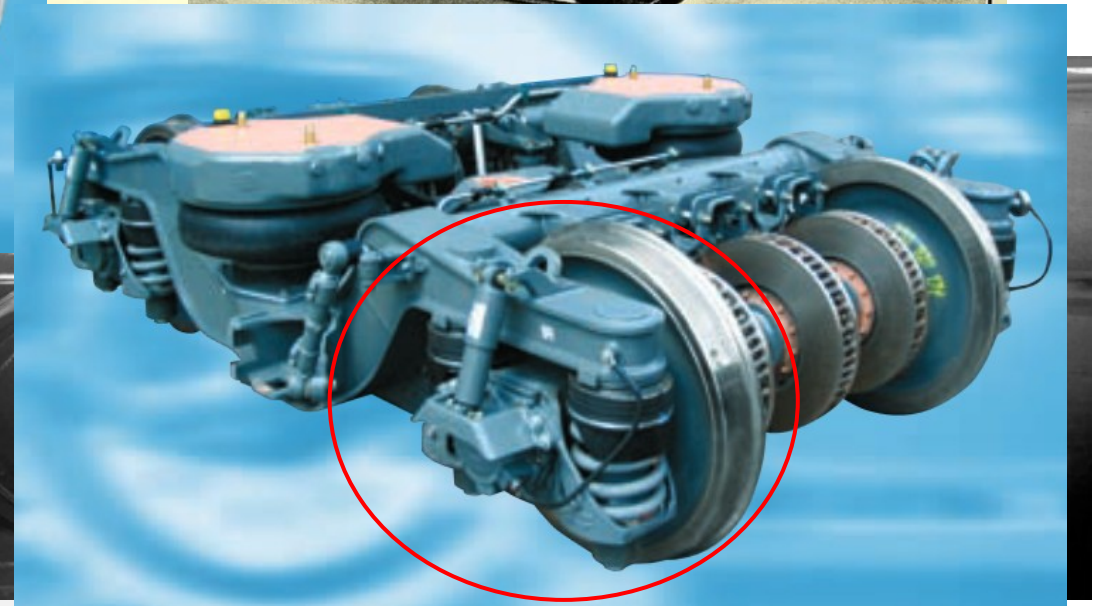
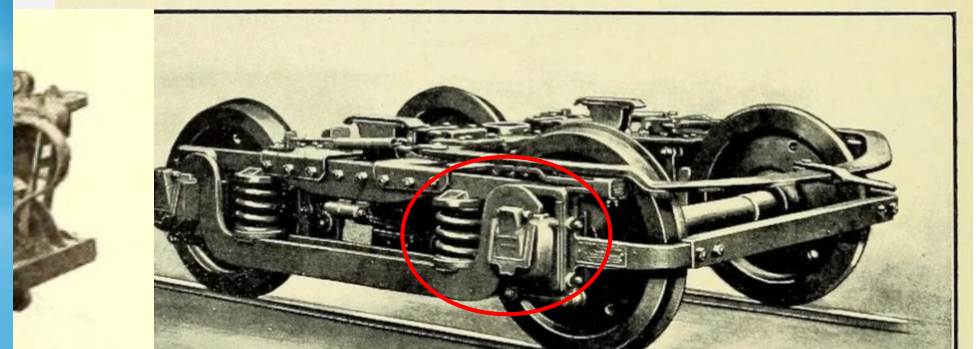
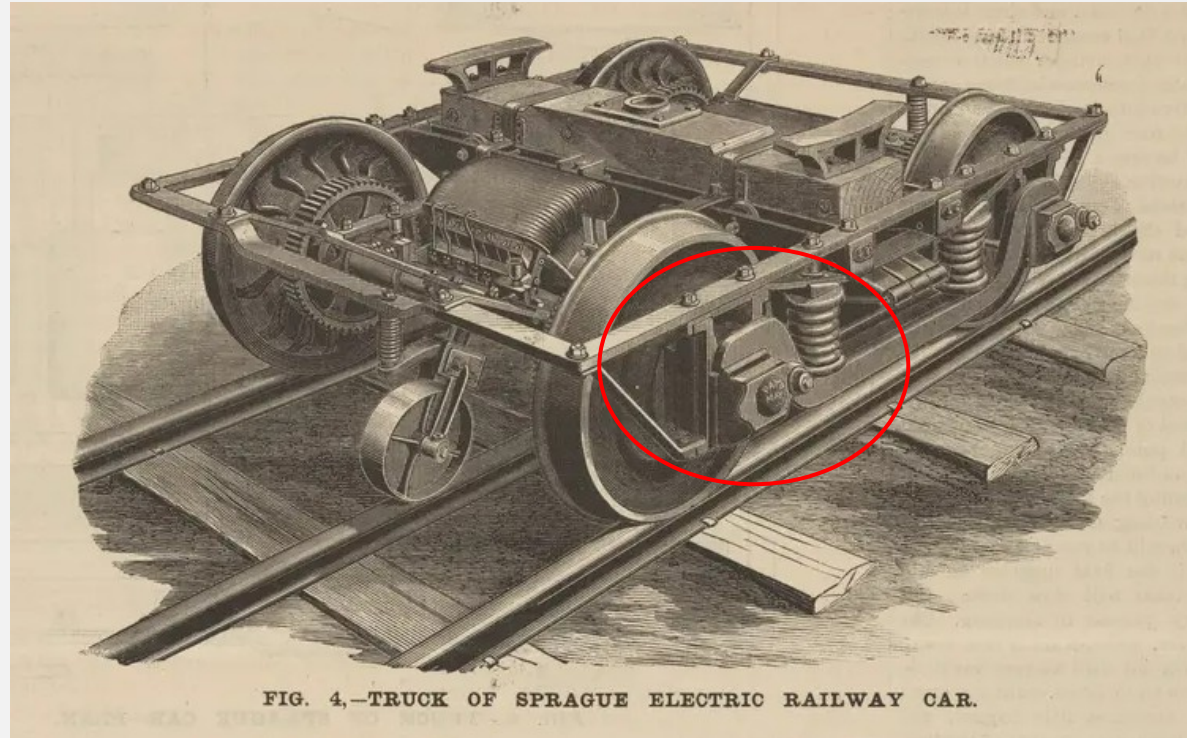


The 'bogie' (or truck) is the interface between the car and the rails

- *Rotate relative to the car*
- *Transfer forces from the rails to the car*
- *Allow the wheelsets to steer through the curves*



Passenger Car Trucks and Bogies



Suspensions

Passenger Car Suspensions

- **Why is there a suspension?**
 - **Vibration Isolation:** Passenger comfort, prevent damage to goods and to the car itself.
 - **Load equalization:** transfer the car weight evenly to the rails.
 - **Control vehicle motion:**
 - curves, switches, crossovers, etc.
 - traction and braking
 - lateral curving forces
 - dynamic stability



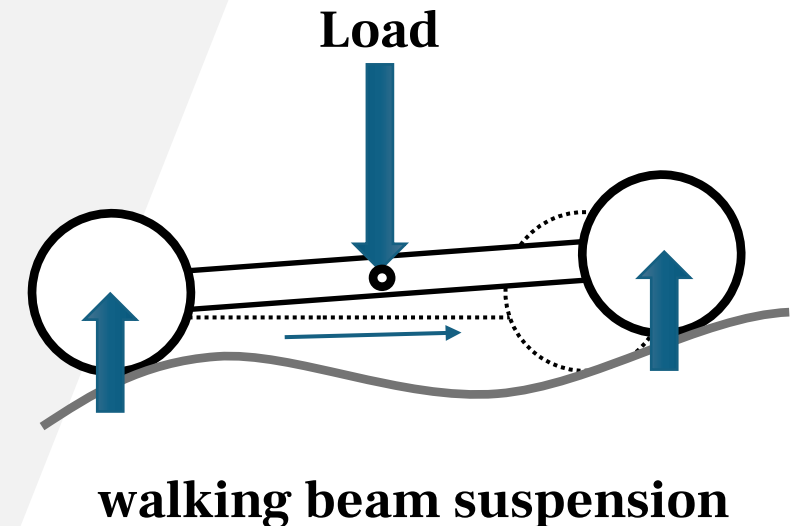
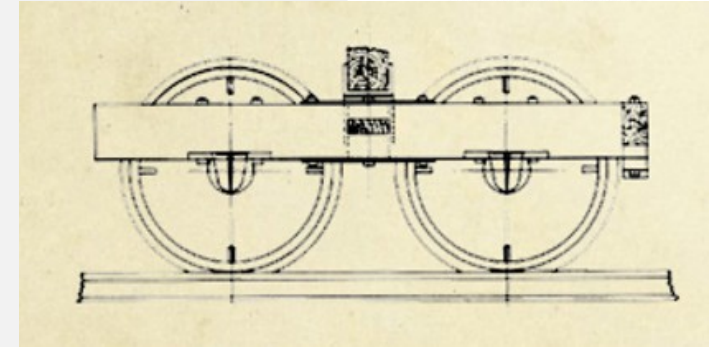
Load Equalization

- 1. Transfer the car weight evenly to the rails.**
- 2. A car with no suspension would lift wheels on even slightly uneven track, such as when entering curves.**
- 3. Rail and subgrade designs place a limit on the maximum wheel load on the rail.**

Load Equalization

A simple way to spread the vertical load to several wheels and keep the loads equalized is to use a walking beam suspension

- The bogie balances the load between the front and rear wheels by pivoting at the bolster.
- The wheel load is always equally split!
- Not as effective at higher speeds...



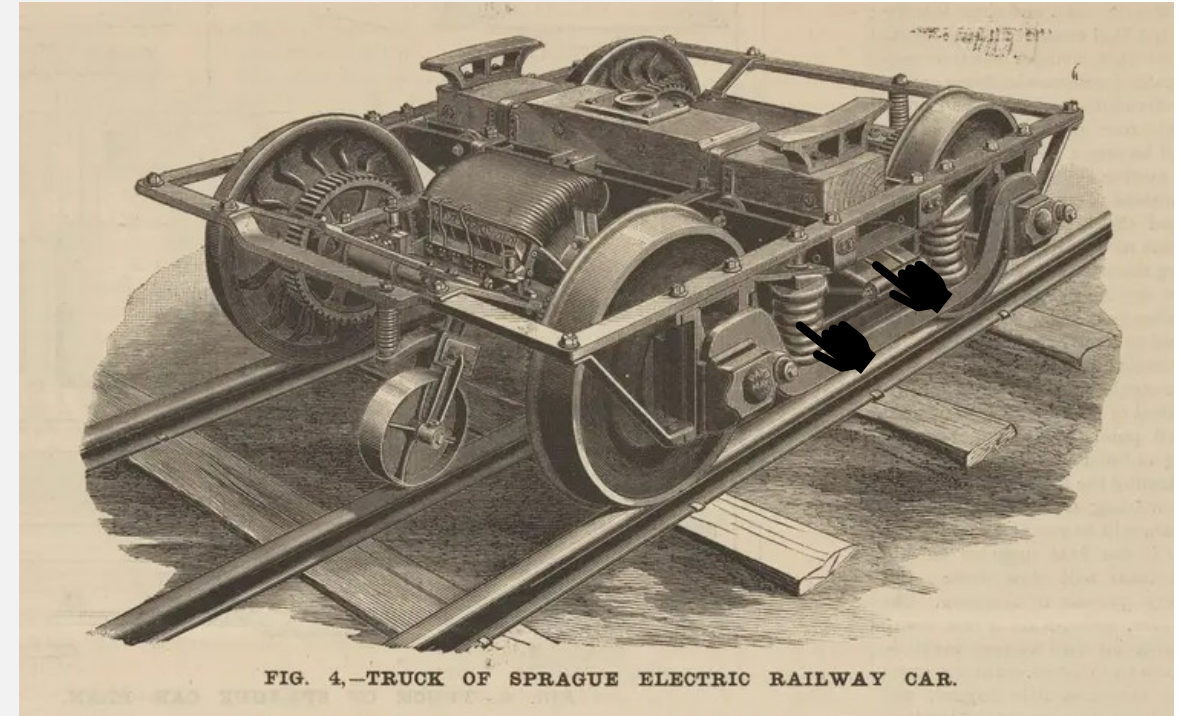
Load Equalization

How to improve this?

- Allow the wheels to move relative to the truck
- Allow the truck to move relative to the car body

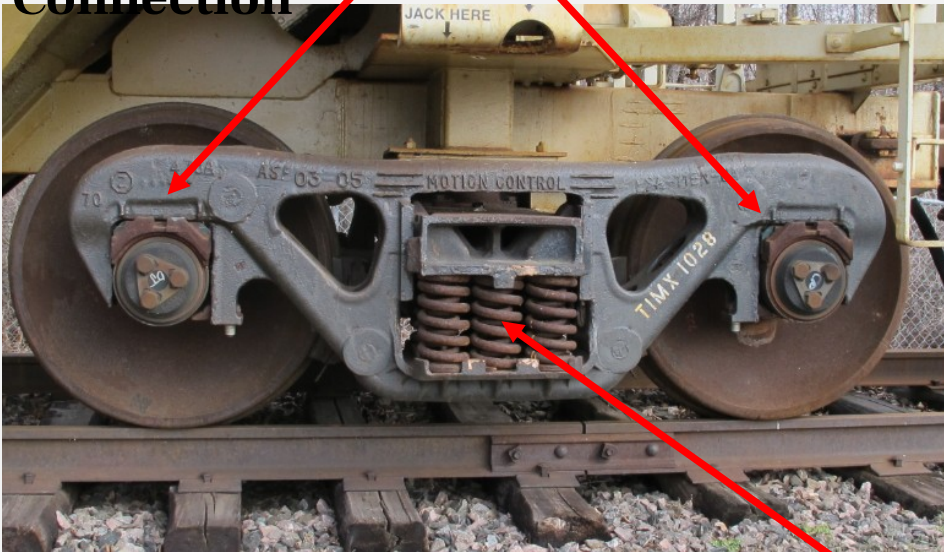
Primary Suspension

Secondary Suspension



Primary and Secondary Suspension

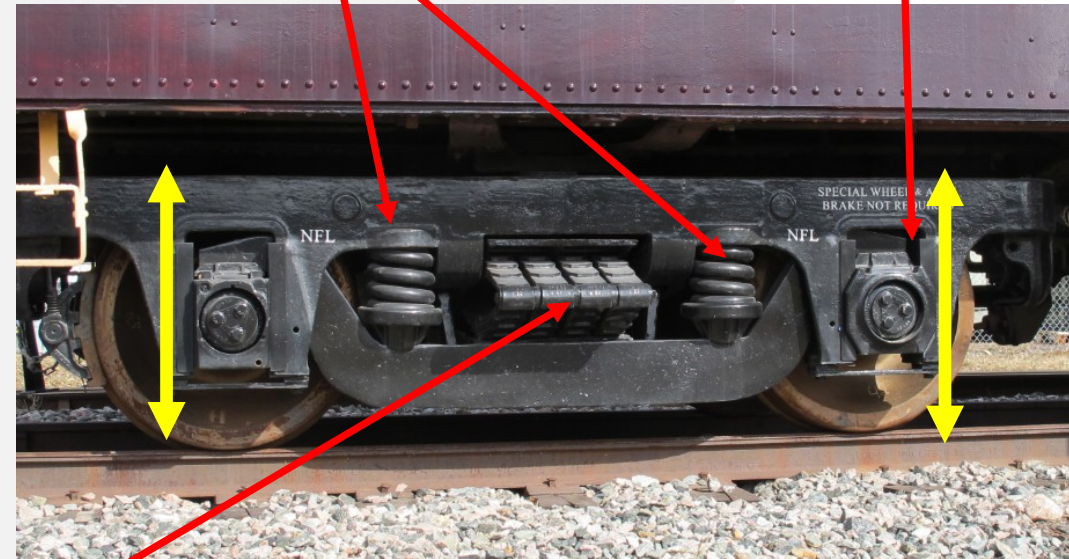
No Primary Springs i.e. “Hard”
Connection



Freight Truck

Primary springs

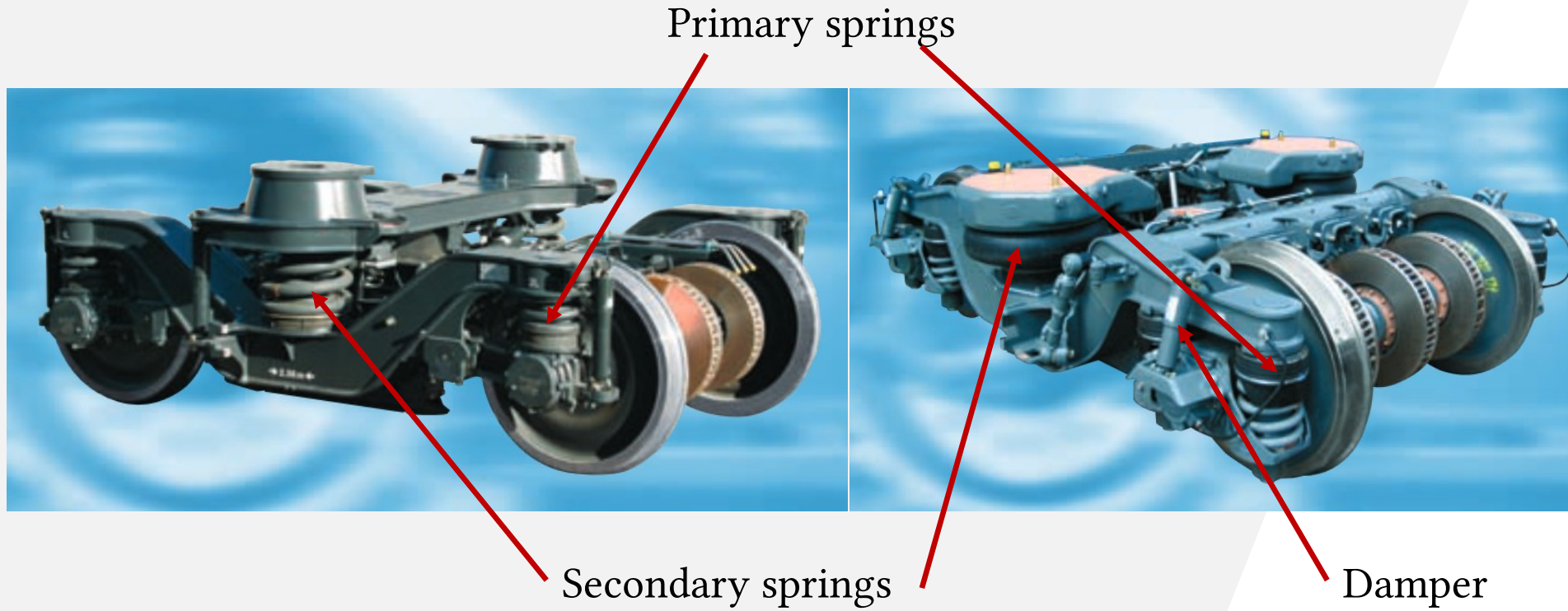
Vertical freedom



Passenger Truck

Secondary springs

Passenger Car Truck



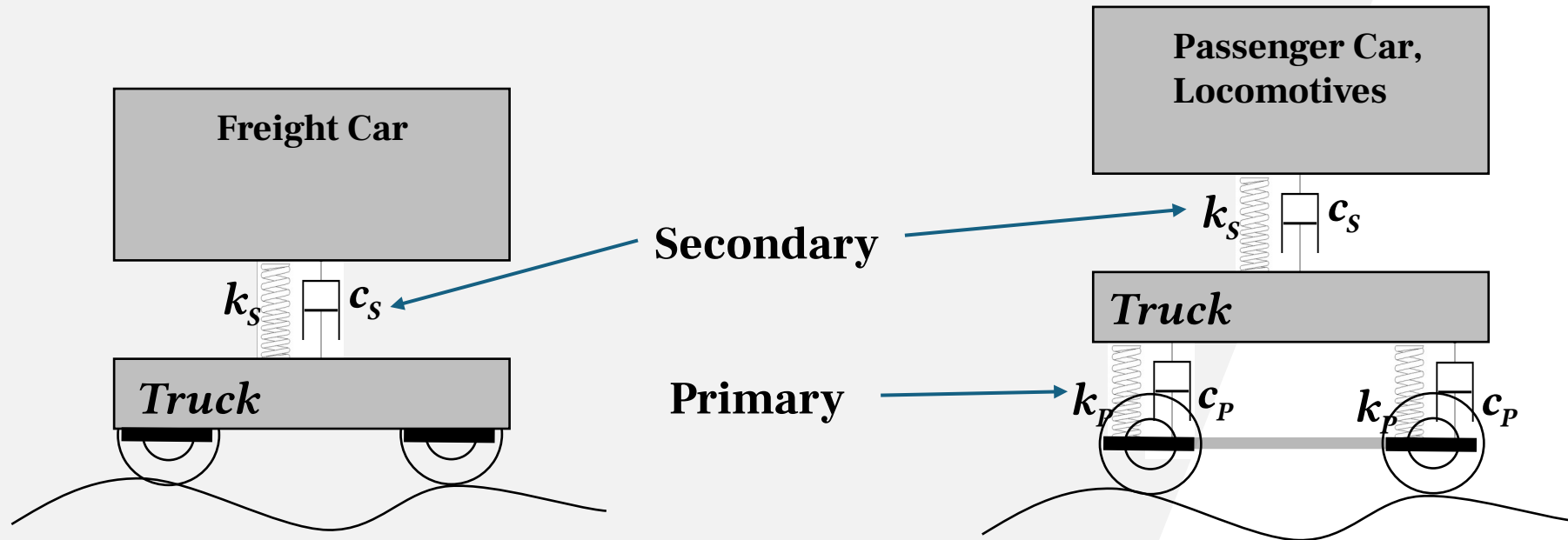
<https://www.mobility.siemens.com/mobility/global/SiteCollectionDocuments/en/rail-solutions/components-and-systems/bogies-catalog-en.pdf>

Primary and Secondary Suspension

- 1. Primary Suspension: A suspension at a wheel location, between a wheelset and the truck frame.**
- 2. Secondary Suspension: A suspension between a car body and a truck frame.**

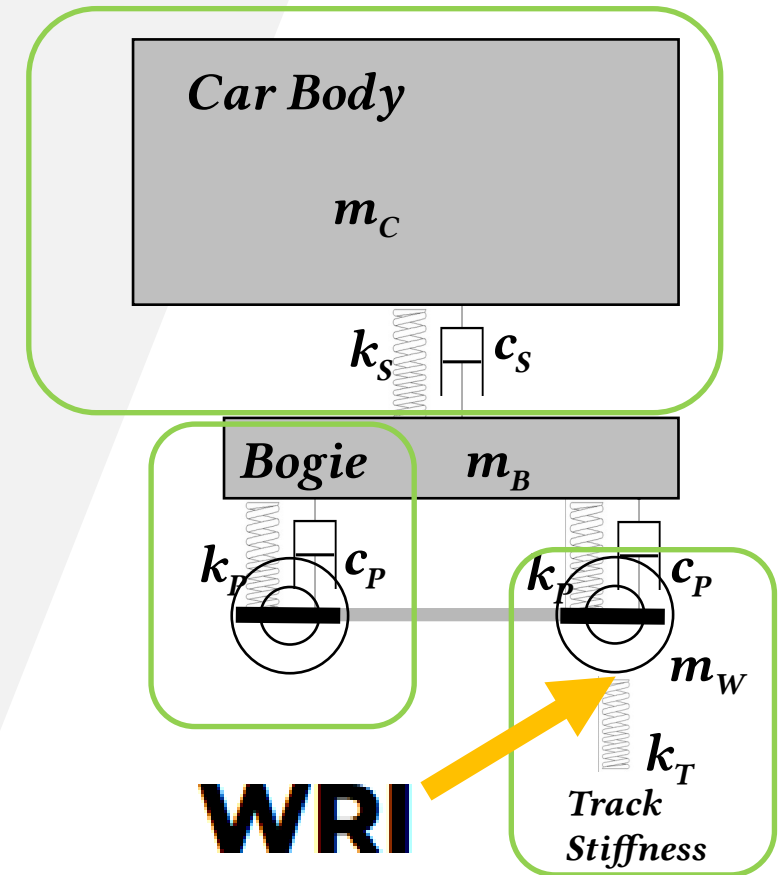
Primary and Secondary Suspension

A basic analysis can be done using a “Quarter Car” model of a rail vehicle



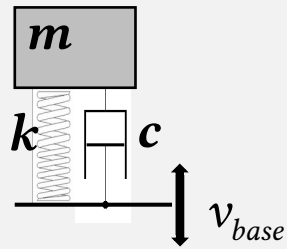
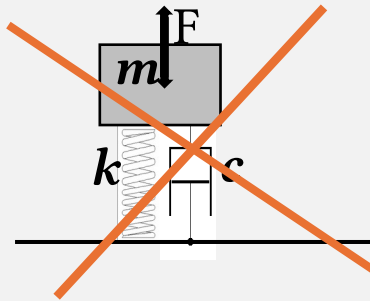
Spring-Mass-Damper System

- **For rail cars**
 - **Main mass:** car body
 - **Secondary Suspension:** between the car body and the bogie
 - **Primary Suspension:** between the wheelset and the bogie

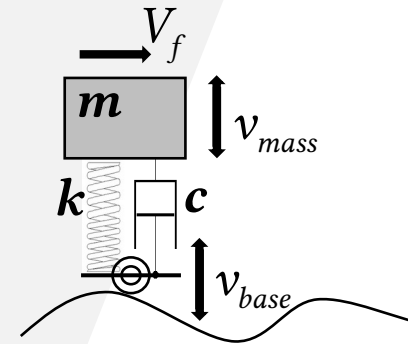




Spring-Mass-Damper System



*“Quarter
Car Model”*

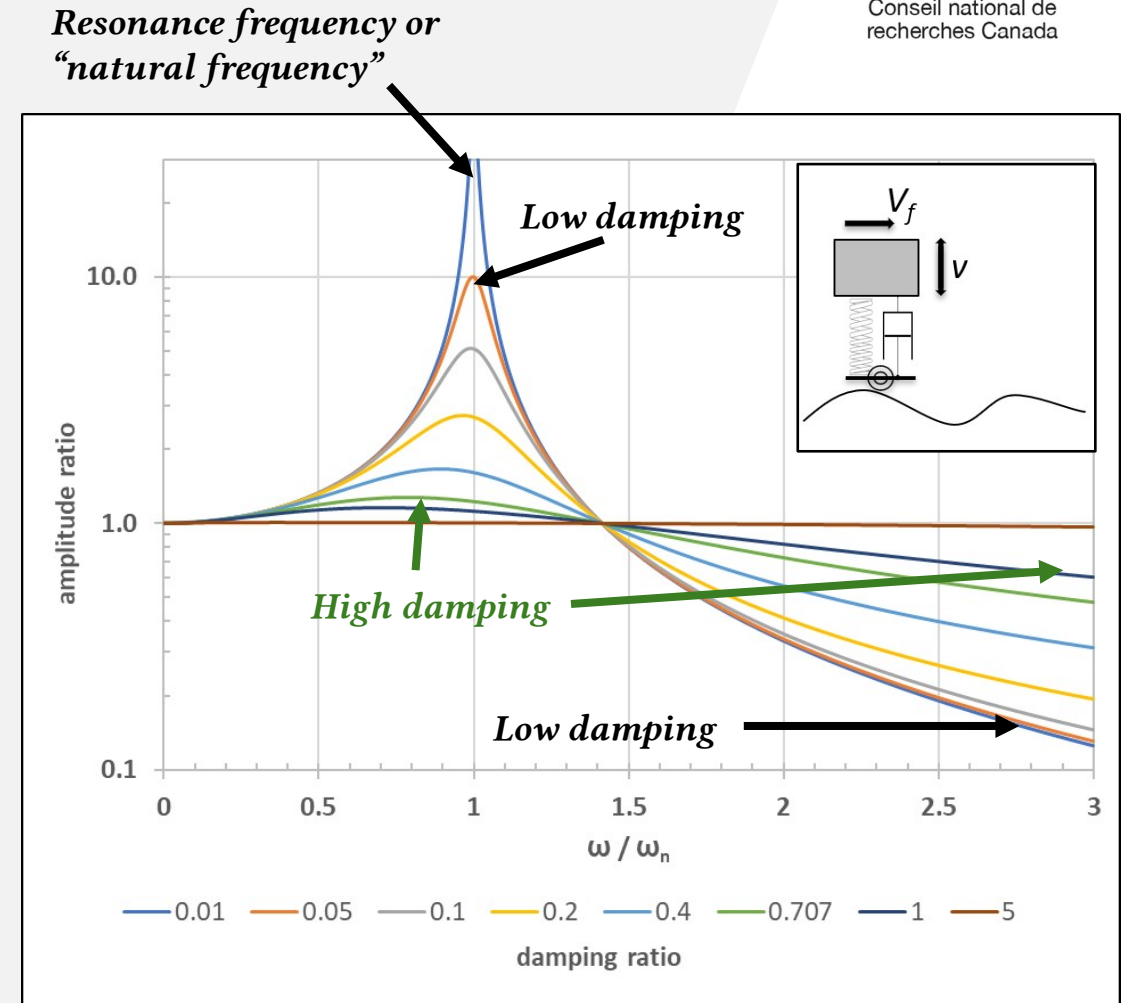


$$\omega_n = \sqrt{k/m}$$

Vibration Isolation

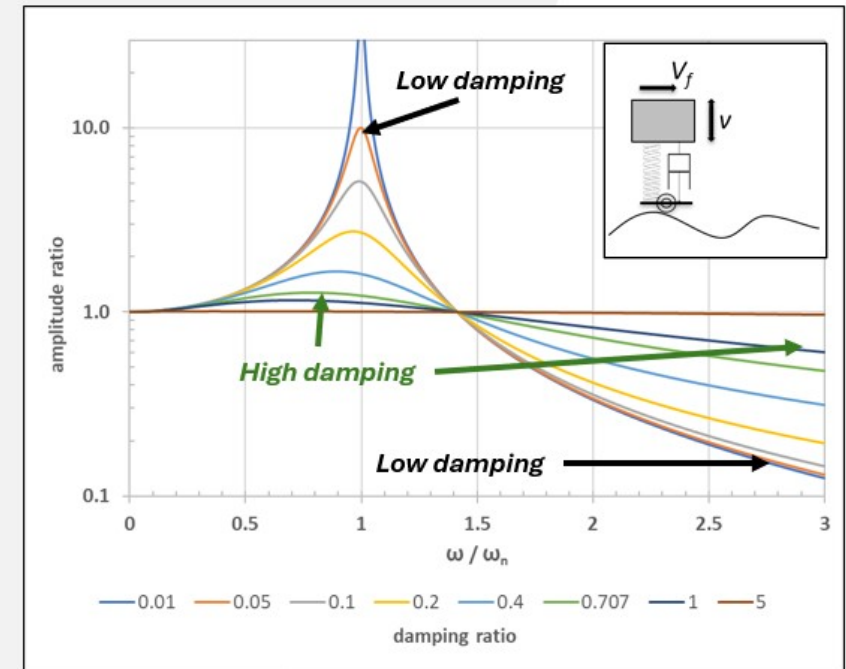
Spring-mass-damper systems have well understood properties:

- Resonance frequency.
- Damping has an effect on the system response.

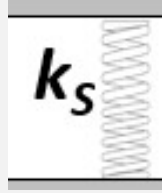


Spring-Mass-Damper System

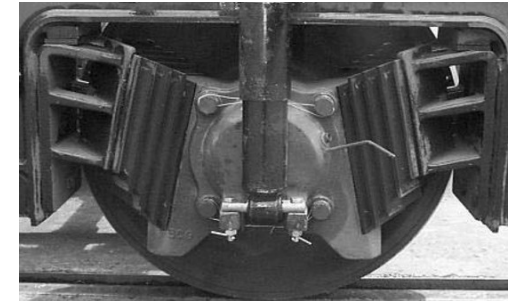
- At the resonance frequency, the mass moves at a very high displacement.
- This can be controlled with damping.
- Above resonance, the mass is isolated from the vibrations applied to the spring.
- Less damping = high displacement at resonance.
- More damping = less isolation at all inputs!



Springs

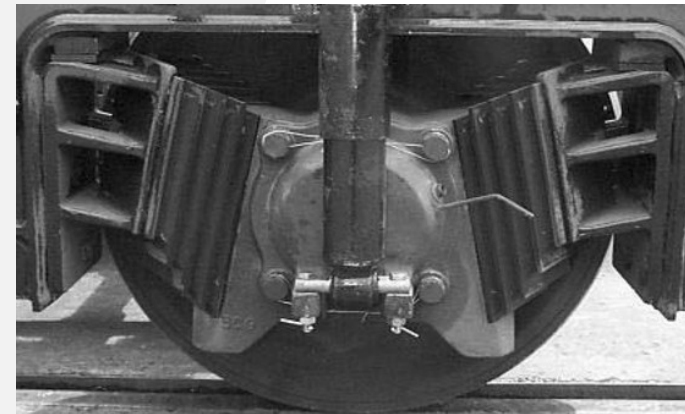
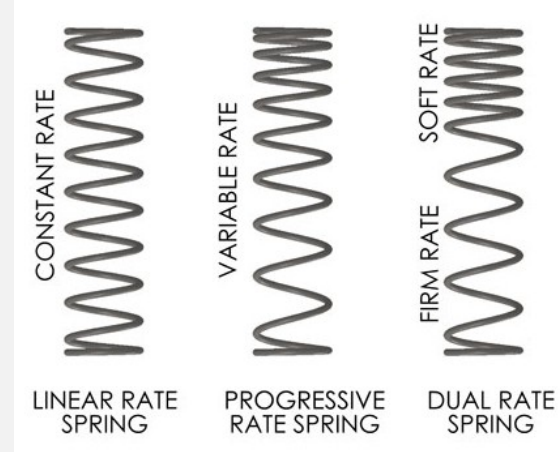


- A spring is anything that produces and increasing force with an increase in displacement
- coil spring and leaf spring are most familiar to most people (seen on ours cars and trucks)
- air springs, rubber/polymer springs



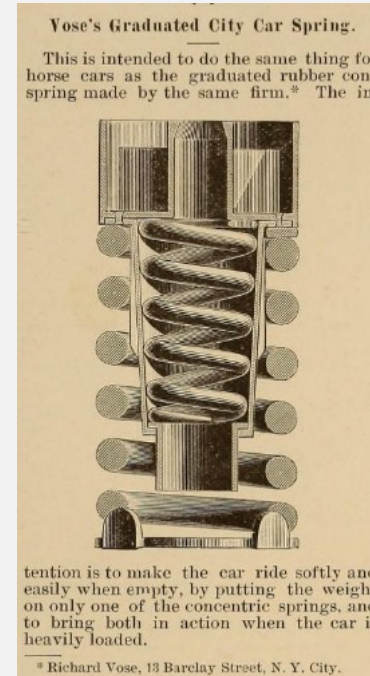
Springs

- **Progressively wound wire springs: become stiffer as the coil is compressed**
- **Dual rate: two wrap designs on one spring**
- **Elastomeric springs (rubber):**
 - Wedge shaped, or cone shaped
 - It's a spring, with some damping
 - Act in 3 directions with 3 different stiffnesses!

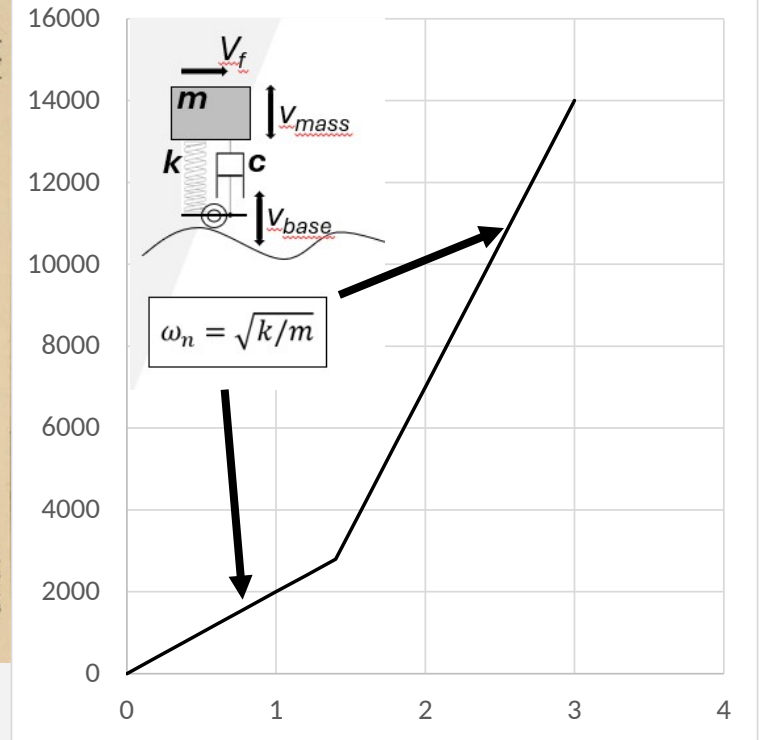


Springs

- How to handle difference between empty car and loaded car?
- Nested coil springs:
- Piece-wise-linear spring rate: lower spring rate when empty, higher spring rate when loaded.
- Leaf, air, and rubber springs also can be designed to be progressive rate (non-linear)



Nov 1884



Air Springs

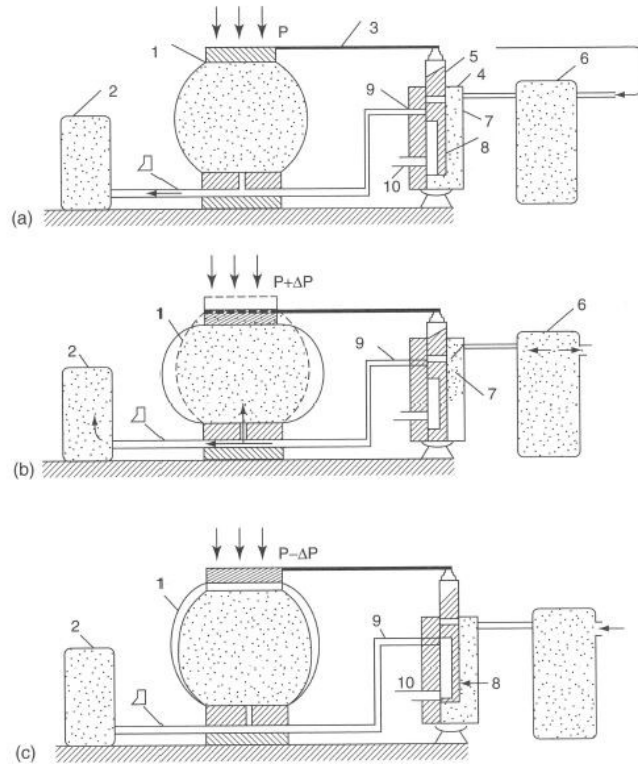


FIGURE 3.11 Schematic showing the operation of a typical air suspension: (a) Equilibrium position; (b) Upstroke; (c) Downstroke.

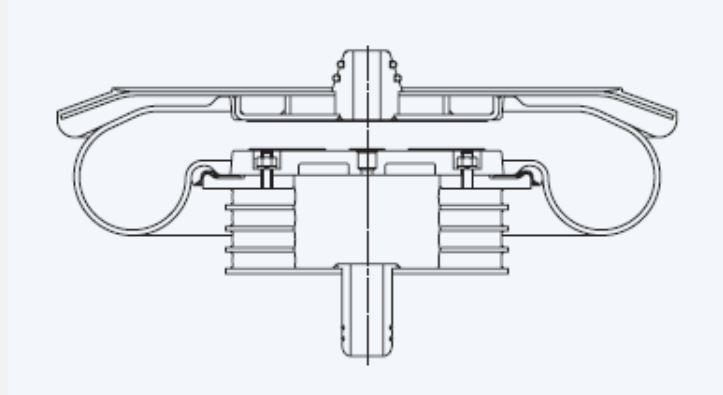
Handbook of Railway Vehicle Dynamics. Iwnicki, 2006, CRC Press

- **Contained gas can act as a spring, but with interesting properties!**
- **The stiffness is a function of the volume and pressure of the contained gas, and the area of the rigid load support.**
- **Very good at high frequency (acoustic) vibration isolation;**
 - **no rigid connection between mass and base**

Air Springs

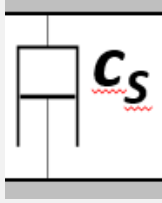


<https://www.bridgestoneindustrial.eu/downloads/Air-Spring-EN.pdf>

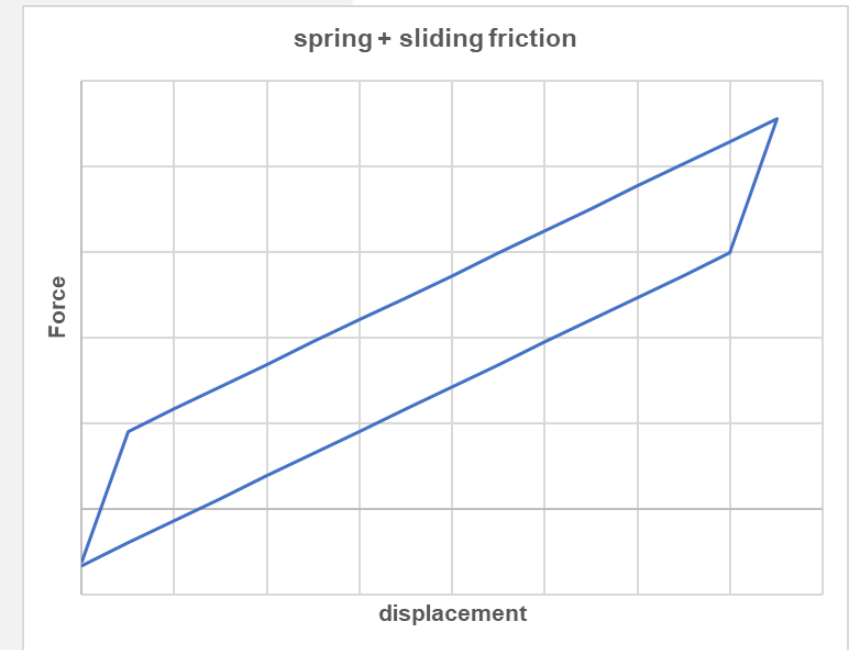


- **Air Springs properties:**
 - **Low lateral stiffness, especially when compared to coil or leaf springs.**
 - **The empty and loaded stiffness can be adjusted.**
 - **Pre-loading the system allows for constant ride height and load levelling.**
 - **Suspension stations can be connected and interact:**
 - **Automated levelling; roll in curves**
 - **controlled pitch and roll response by controlling air flow, and using orifices and reservoirs**

Damping

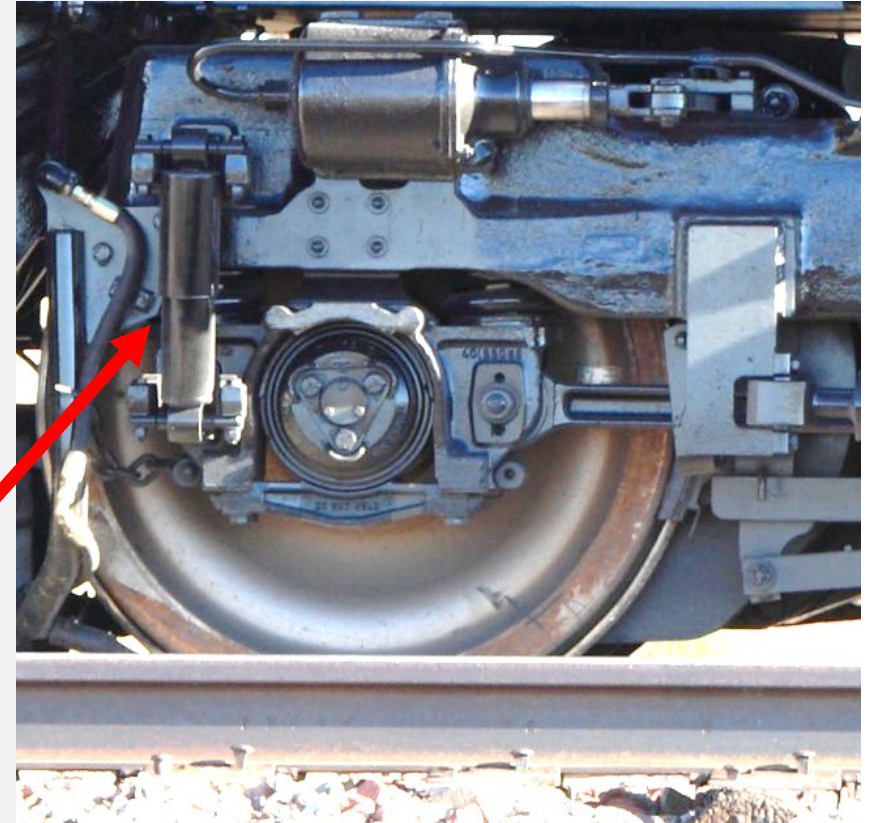
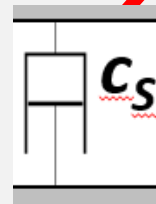


- What is “Damping”?
- Energy loss in the system, through friction.
- In an automobile, typically an oil-filled “shock absorber”
- Pushing the oil through a hole involves friction
- But any form of friction works – metal-on-metal sliding can be used as a form of damping



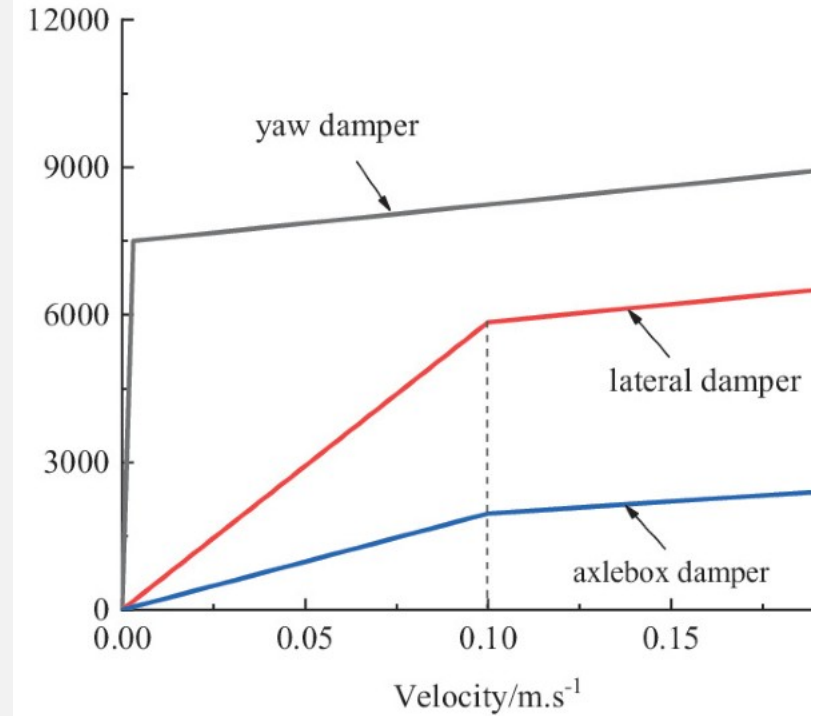
Damping

- Freight rail cars almost universally use surface friction for damping; “Friction Wedges” in the truck design.
- Hydraulic dampers are often seen on locomotives and passenger rail car trucks.



Hydraulic Dampers

- Force is proportional to velocity.
- Design details can be adjusted to result in non-linear response to velocity, include “blow off” force limits, and other useful design features.



Ren, J., Li, Q., Ren, Z., Yang, G., Wu, Y. (2025). The Availability of Four Kinds of Hydraulic Damper Models on the Vehicle Dynamics Investigation. In: Huang, W., Ahmadian, M. (eds) Advances in Dynamics of Vehicles on Roads and Tracks III. IAVSD 2023

Bogie Suspensions: overview

- **Load equalization:**
 - allow tractive effort to get to the rails
 - Keep loads distributed evenly, uneven loads could overload the rail
- **Vibration isolation:**
 - Isolate passengers from rail disturbances.
- **Allow the car to maneuver through all the twists and turns of the track system**
 - Negotiate curves and special trackwork!

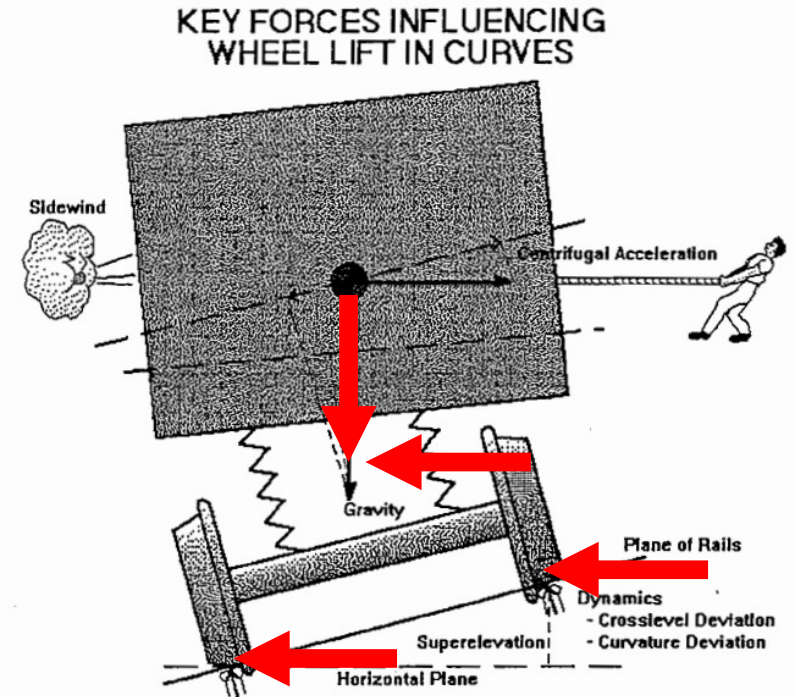
Passenger Car Curving

- **Curves:** lateral loads on the car body are balancing centrifugal forces with gravity and wheel-rail forces to make the car “go around the curve”.
- **What on the car resists or assists these forces?**
 - **Bogie to car body:** Center plate, side bearings
 - **Bogie to rail:** wheels, bogie frame, bolster, springs

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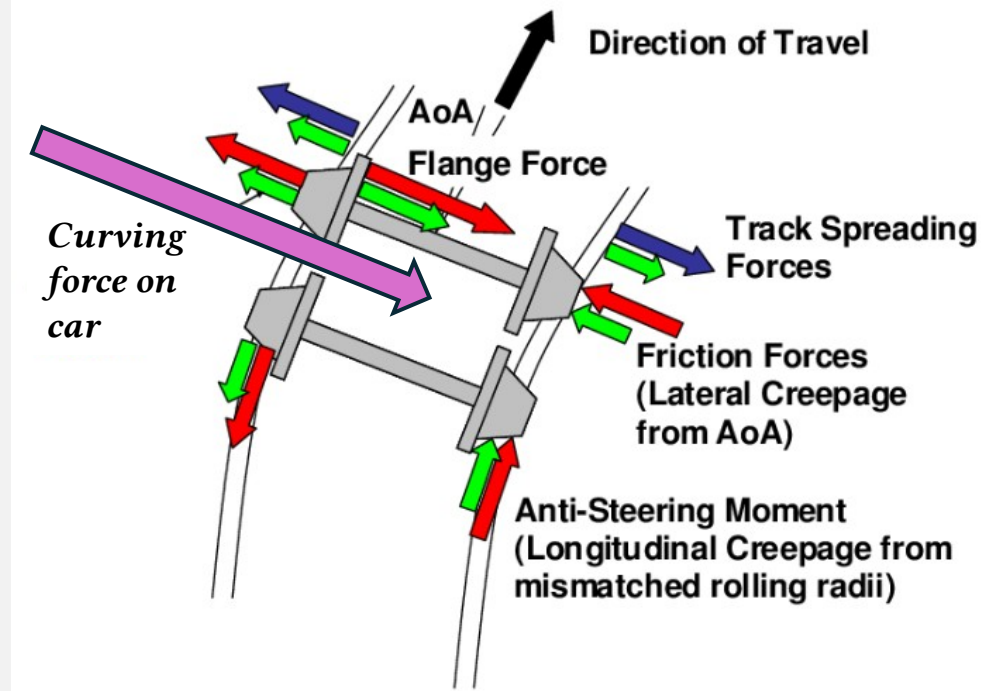
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High Cant Deficiency Operation of the Talgo Train on the Pacific Northwest Corridor, ENSCO 1999.

Passenger Car Curving

- How do you get solid axle wheelsets, mounted to a rigid frame, to go around a curve?
- The bogie frame must allow some relative motion of the wheelsets, either with controlled stiffness, or clearances.
- Where do all the forces go?
(it's complicated....
...but they always add up to push the car around the curve.)

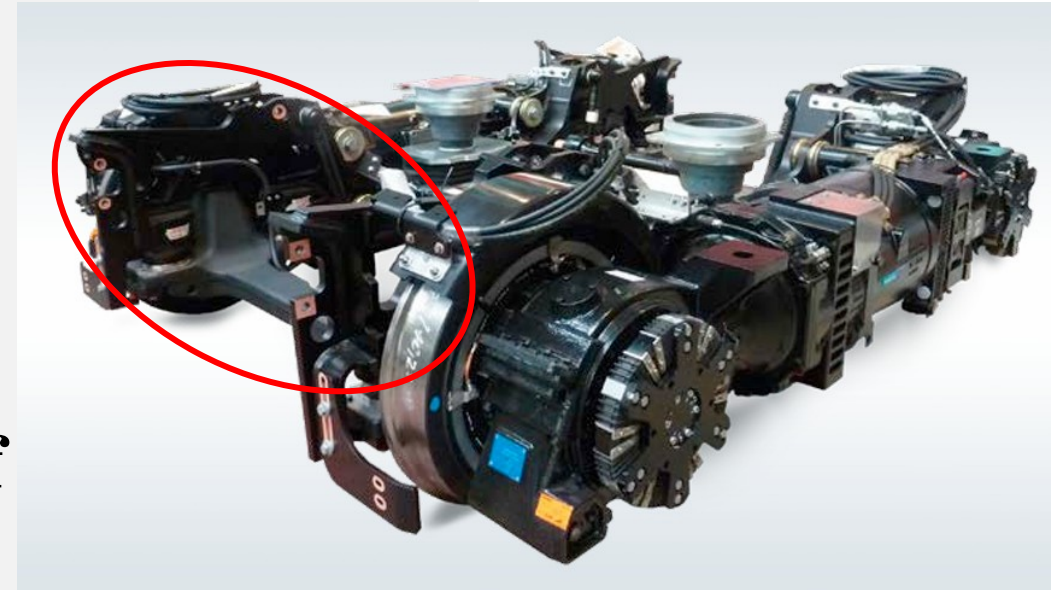


Interaction between TOR / wheel tread friction levels and track spreading forces

Top of Rail Friction Control as a Means to Mitigate Damaging Lateral Loads Due to Overbalanced Operation of Heavy Axle Load Freight Traffic in Shared High Speed Rail Corridors. Oldknow, Kevin; Eadie, Donald; ASME JRC 2010

Passenger Car Curving

- **For trucks with rigid axle wheelsets, there are limits to the curve radius a wheelset can safely negotiate around a curve**
 - controlled by wheel-rail interaction limits
- **What if the axle wasn't there? What if the wheels were independent?**
 - Longitudinal forces are reduced to those of drive/brake traction



https://assets.new.siemens.com/siemens/assets/api/uuid:67afff1c-e169-4470-a6a1-bf105db846b1/mors_b10_025-00-datasheet-bogies-sf35-deenus-144_original.pdf

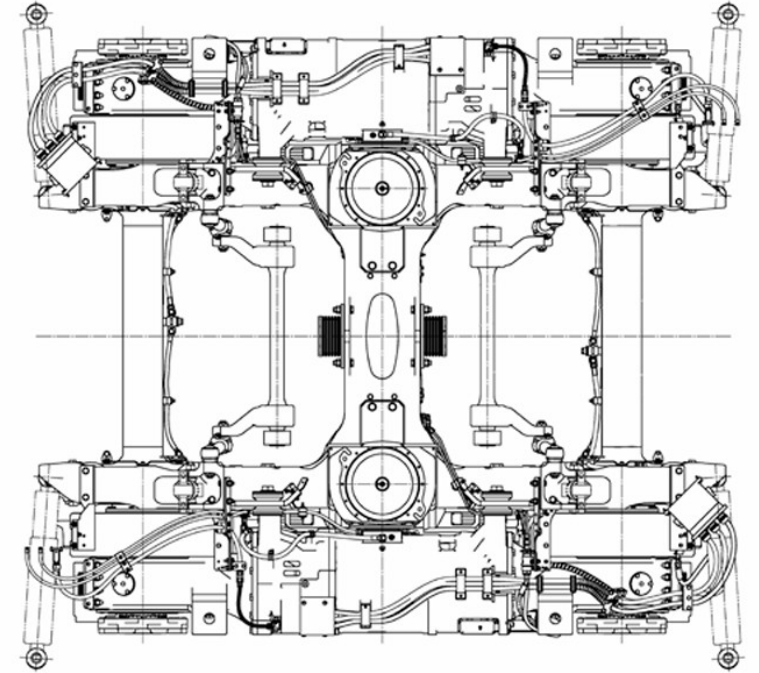
Passenger Car Curving: IRW

- **Why aren't all trains like this!**
- **Trade-offs...**
 - **Steering effect of tapered rigid axle wheelset is lost, requires design changes.**
 - **Unique wheel-rail interaction effects = new types of wear/noise mechanisms.**
 - **Speed limitations**
 - **Requires truck frame and wheel/bearing interface to be designed for the new load conditions.**
 - **The loss of the axle transfers bending moment loads to the bearings, and truck frame**

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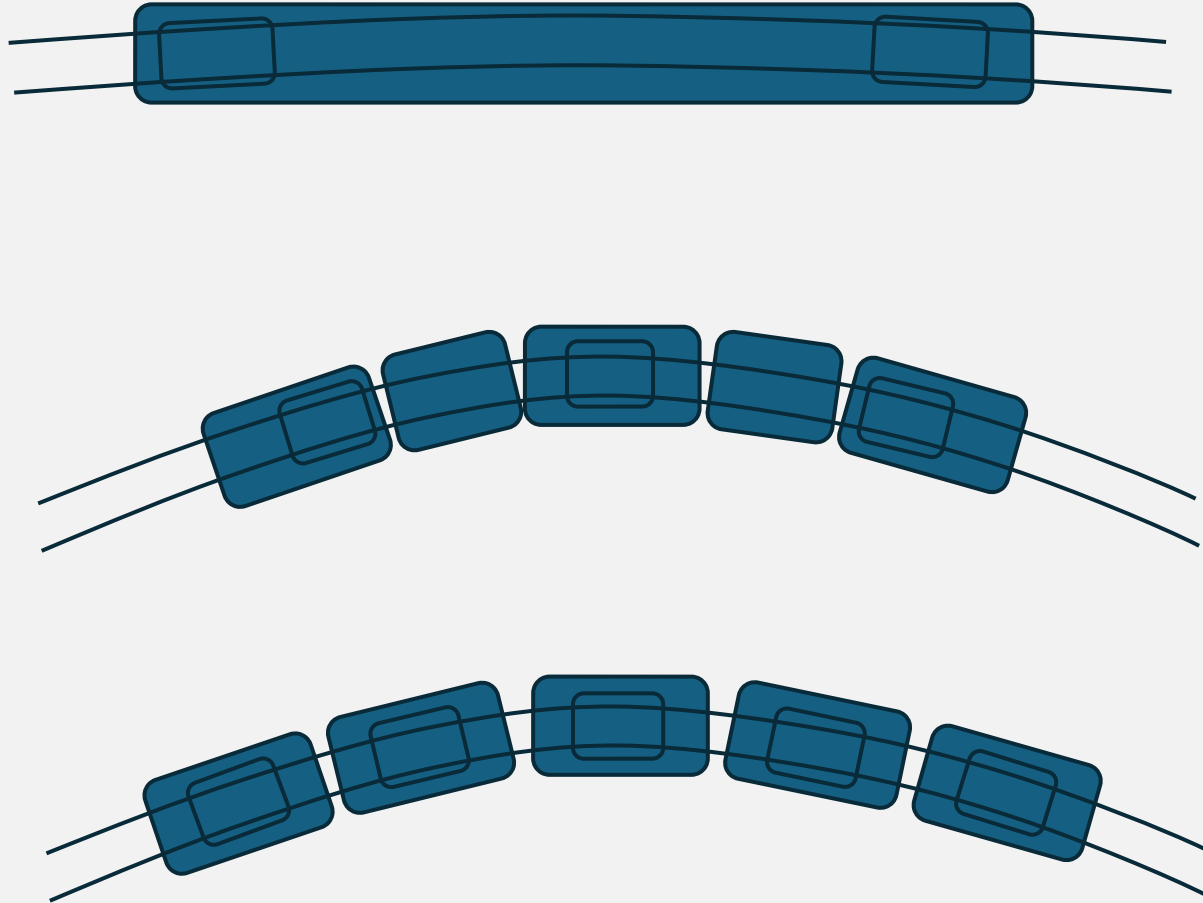
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Passenger Car Curving: IRW



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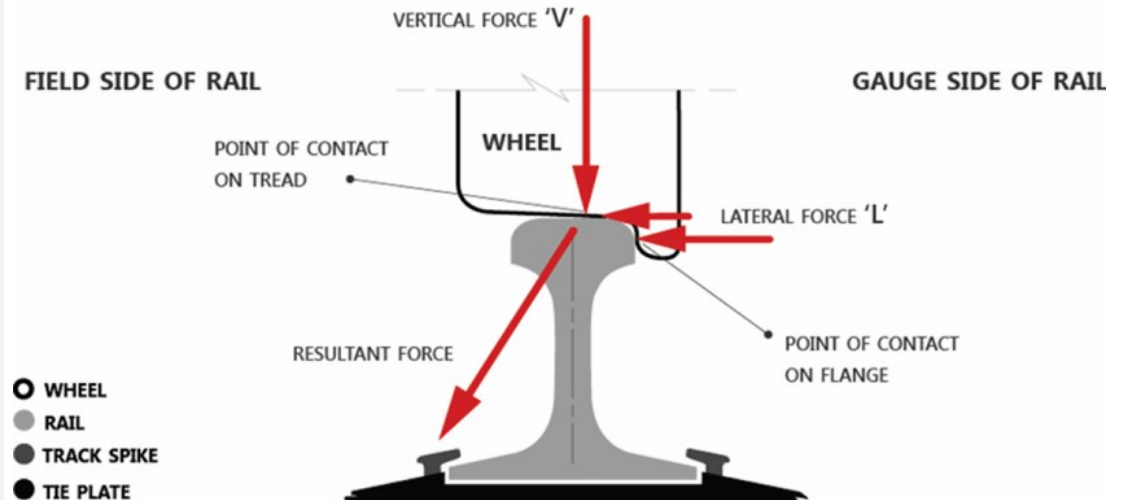
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But what about lateral loads?

...they don't go away!

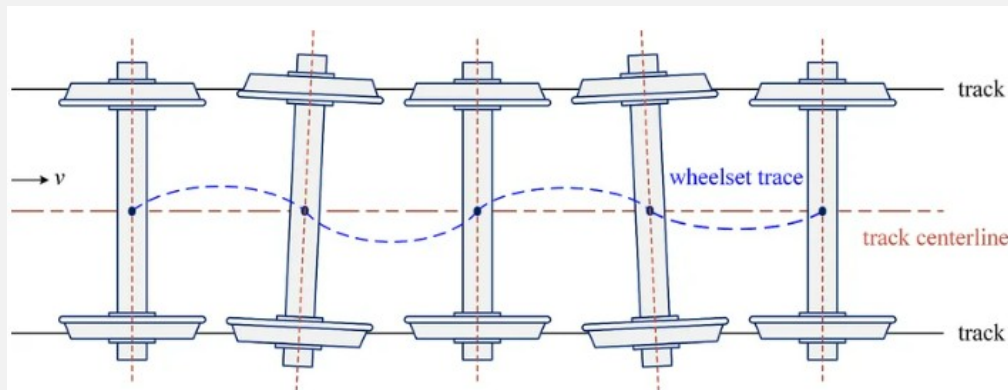
- Flanging during curving
 - check rail



<https://www.tsb.gc.ca/sites/default/files/rapports-reports/rail/R16E0102/eng/r16e0102.pdf>

Passenger Car Dynamic Stability

- Side effect of being able to negotiate curves is that lateral stability is affected.
- Wheel conicity allows a wheelset to roll around a curve:
 - more conicity = tighter curves
- But more conicity = greater self-centering forces.
- At high speeds the truck may begin “hunting”.



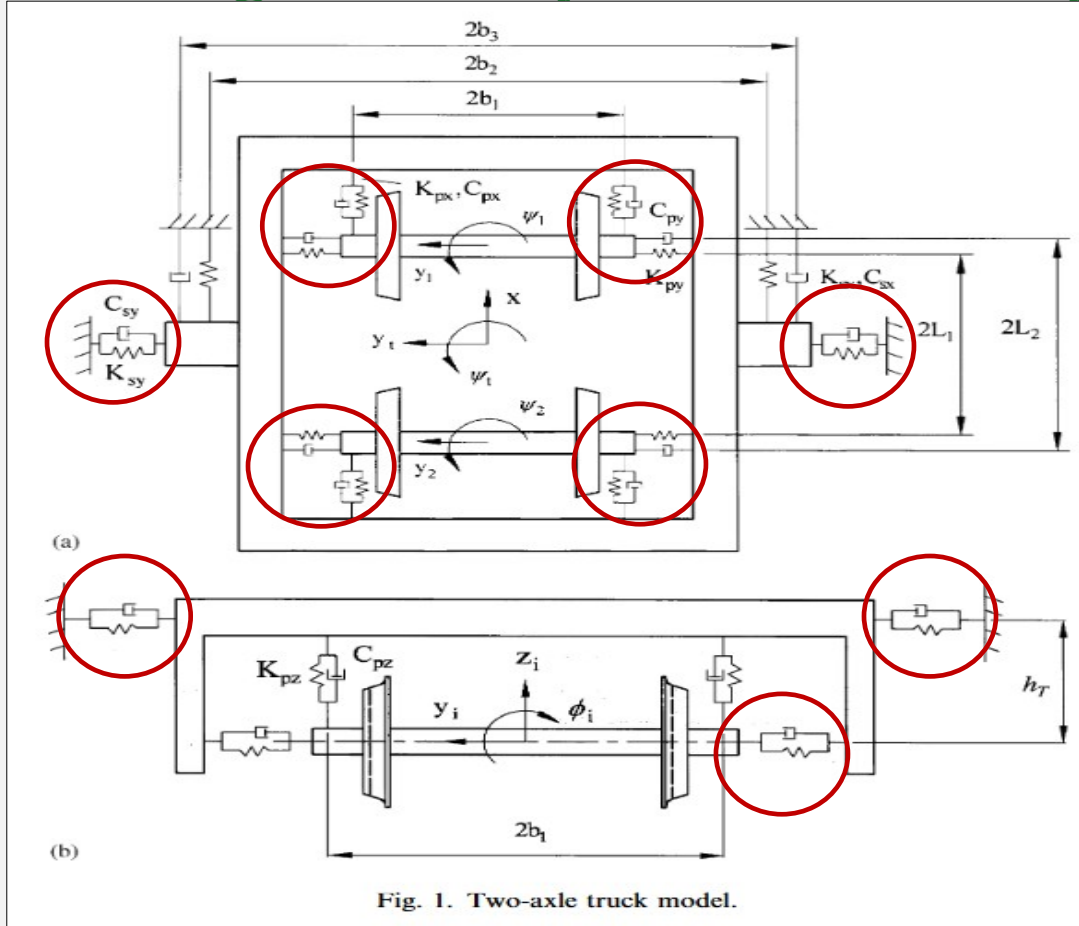
Wang, Jiacheng & Ling, Liang & Wang, Kaiyun & Zhai, Wanming. (2024). Nonlinear stability evolution of railway wagon system due to wheel profile wear. Nonlinear Dynamics. 112. 1-21. 10.1007/s11071-024-09665-x.

Passenger Car Dynamic Stability

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Journal of Sound and Vibration 282 (2005) 881–898, Hunting stability analysis of high-speed railway vehicle trucks on tangent tracks. Sen-Yung Lee, Yung-Chang Cheng



Bogie Suspensions: overview

- Load equalization:
 - Equal load for all wheels
 - allow tractive effort to get to the rails
- Vibration isolation:
 - Keep rail disturbance from causing the car body to 'feel' every bump in the rail
- Allow the car to maneuver through all the twists and turns of the track system
- Dynamic Stability
 - no uncontrolled oscillations at high speeds!



Thank You!

Questions?

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